

Pratham P Shetty

Backend Software Engineer — Distributed Systems — AI/ML Engineer

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Visa Status: Eligible for Overseas Visa Sponsorship and Relocation



PROFESSIONAL SUMMARY

Backend-focused Software Engineer with hands-on experience designing scalable APIs, distributed systems, workflow automation platforms, and production-grade backend infrastructure. Skilled in FastAPI, Nest.js, Django, PostgreSQL, Docker, RabbitMQ, LangGraph, and cloud-native deployments, with proven success improving API performance by up to 40% and building reliable orchestration workflows for large-scale automation systems. Experienced in backend architecture, database schema design, real-time systems, DevOps workflows, and scalable production deployments, with strong interest in Golang, distributed systems, fintech infrastructure, and workflow orchestration platforms.

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript, TypeScript, Dart, Kotlin, C#, Golang

Backend Frameworks: FastAPI, Django, Flask, NestJS, Express.js

Frontend & Mobile: React, Next.js, Angular, Flutter, Jetpack Compose

Databases: PostgreSQL, MongoDB, Firebase, SQLite, SQL

Distributed Systems & Messaging: RabbitMQ, WebSockets, Workflow Automation, Event-Driven Systems

DevOps & Infrastructure: Docker, Nginx, GitHub Actions, Swagger, Postman, Azure VPS, CI/CD

AI/ML: LangChain, LangGraph, RAG, YOLOv8, OpenCV

PROFESSIONAL EXPERIENCE

Software Developer — DevAppSys IT Solutions [P] Ltd.

Jan 2026 – Present

Mangalore, India

- Engineered scalable backend APIs and distributed backend systems using FastAPI and Node.js for production-grade deployments.
- Designed optimized PostgreSQL database schemas and backend workflows for enterprise applications.
- Optimized backend services and reduced API response latency by **40%** within 6 months of deployment.
- Integrated Dockerized CI/CD deployment workflows improving release reliability, scalability, and operational efficiency.
- Participated in backend architecture discussions, API documentation, debugging, and production reliability improvements.

Research Intern — NITK Surathkal

Jan 2025 – Sep 2025

IoT & Computer Vision Research

Karnataka, India

- Researched IoT-enabled computer vision systems for intelligent monitoring and automation workflows.
- Optimized YOLOv8 object detection pipelines, improving real-time inference performance by **15%**.
- Converted PyTorch models into ONNX format for efficient deployment on edge and embedded devices.
- Integrated AI-powered vision workflows with IoT systems under faculty-led research initiatives.

- Developed responsive React interfaces and RBAC-based authentication systems for enterprise web platforms.
- Scaled backend APIs using FastAPI and Django for high-performance production environments.
- Collaborated with teams of 5+ developers to improve deployment reliability and software quality.

PROJECTS

VTU Diary - AI Automation Platform

FastAPI, React 19, LangGraph, Groq, PostgreSQL, Docker

Full Stack AI Automation Platform

- Architected and deployed a scalable AI-powered workflow automation platform serving **1,500+ registered users**.
- Designed distributed workflow orchestration pipelines using LangGraph for automated VTU diary processing and submission workflows.
- Engineered PostgreSQL relational schemas and optimized backend data models for scalable workflow execution.
- Built fault-tolerant backend services with resumable multi-day workflow recovery and duplicate prevention mechanisms.
- Implemented APScheduler-based automated background workflows running daily at **15:55 IST**.
- Integrated Groq LLM APIs and reusable authenticated session management for automated content generation pipelines.
- Built modular FastAPI backend architecture with isolated services for users, logs, administration, and workflow execution.
- Containerized infrastructure using Docker Compose with FastAPI, PostgreSQL, React, pgAdmin, and Nginx services.
- Achieved **7,273+ Google search impressions**, **423 organic clicks**, and ranked **#3 on Google**.
- Live Platform: autodiary.tech

Distributed Workflow Queue System

FastAPI, RabbitMQ, PostgreSQL, Docker

Backend Distributed Systems Project

- Developed a distributed task processing system using RabbitMQ and FastAPI for asynchronous workflow execution.
- Implemented retry queues, task acknowledgements, and failure recovery mechanisms for reliable background processing.
- Designed PostgreSQL schemas for workflow tracking, execution logs, and distributed task state management.
- Built scalable worker-based architecture supporting concurrent processing and event-driven backend workflows.
- Containerized microservices using Docker Compose enabling scalable local and cloud-native deployments.

SIT Mangalore App

Flutter, Dart, Appwrite, Firebase FCM

Cross-Platform Mobile Application

- Developed and deployed a cross-platform academic mobile application serving **1,500+ students and faculty**.
- Built Flutter-based mobile architecture using BLoC state management for Android and iOS platforms.
- Integrated Appwrite Backend-as-a-Service for authentication, storage, and real-time synchronization workflows.
- Implemented OAuth2 and JWT-based access control for administrators, students, and faculty members.
- Engineered WebSocket-powered real-time notice delivery eliminating client-side polling requirements.
- Integrated Firebase Cloud Messaging with topic-based notification broadcasting for targeted alerts.
- Designed offline-first workflows using Hive local storage for timetable and notice caching.
- Reduced login time by **30%** and achieved application cold-start times below **2 seconds**.
- Achieved **1,000+ Play Store downloads** and received institutional appreciation during the **2023 Convocation Ceremony**.

Tulunad Diaries

Nest.js, React, MongoDB, Swagger, Nginx, MinIO

Production News and Media Platform

- Engineered and deployed a production-grade news and media platform replacing a legacy WordPress infrastructure.
- Developed scalable Nest.js REST APIs with modular architecture for articles, media management, and editorial workflows.
- Built React SPA interfaces with optimized navigation, SEO-focused metadata, and lazy-loaded article rendering.
- Integrated MongoDB for high-speed article storage and optimized large-scale read/write performance.
- Implemented MinIO object storage for scalable image and video asset management workflows.
- Configured Nginx reverse proxy infrastructure with SSL termination, gzip compression, and static asset caching.
- Improved organic search reach by **40%** through structured SEO optimization and metadata integration.
- Reduced page payload sizes by approximately **70%** using optimized media delivery pipelines and compression strategies.
- Achieved sub-**200ms** API response times using optimized MongoDB indexing and caching mechanisms.
- Live Platform: tulunaddiaries.com

EDUCATION

Srinivas Institute of Technology, Mangalore	2022 – 2026
<i>B.E. in Artificial Intelligence & Machine Learning</i>	CGPA: 8.6
Sri Sathya Sai Loka Seva P.U. College, Alike	2020 – 2022
<i>PUC – PCMB</i>	81.5%
Burooj English Medium High School	2019 – 2020
<i>SSLC – 10th Standard</i>	80.8%

ACHIEVEMENTS

Winner – Srinathon 2024 National Hackathon for AI-enabled platform development.
Top 5 – Blockchain Hackathon at Sahyadri College among 30+ competing teams.
Top 10 – AI Hackathon at RV College and invited to Intel technical workshop.
2nd Place – SPRINTHACK 2025 organized by AI & DS Department at SIT.

CERTIFICATIONS

IoT and Computer Vision Research Internship – NITK Surathkal (2025)
Celonis Virtual Internship Certificate
Backend Engineering and Distributed Systems – Self-Learning and Production Projects

LANGUAGES

English, Kannada, Tulu, Hindi
